

SearchPath: A Visual Reminder of Past Search Activity to Support Students with Dyslexia when Undertaking Exploratory Searches

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Abstract

Students often face complex academic search tasks, which cannot be resolved with a single query and selecting among the top few search results, or a single answer provided by a generative AI interface. Exploratory search process have been proposed as a useful framework for enabling searchers to undertake complex search tasks. In recent years, a variety of interfaces have been developed to support such searching. While such approaches generally focus on enabling searchers to make sense of what has been found and supporting them in managing their search activities, an underlying assumption is that searchers have good reading skills, working memory, and attention. However, these assumptions do not hold for all users; there can be a wide range of neurodiversity among students with respect to these important cognitive functions. Some searchers, such as those with dyslexia, need more support in some or all of these aspects than academic search interfaces offer. Taking inspiration from prior research on multi-workspace search environments, visually representing histories of past search activities, tag-based approaches to saving search results, interactive mechanisms for filtering what has been saved, and accessibility, we have developed a novel approach we call SearchPath. Herein we explain the key features of SearchPath, highlighting how they have been chosen and fine-tuned to serve as interactive memory aids that can help students with dyslexia to maintain orientation during their search journey.

CCS Concepts

• **Human-centered computing** → *Accessibility design and evaluation methods; Information visualization; User interface design*; • **Information systems** → *Search interfaces; Digital libraries and archives*.

Keywords

Academic search, Accessible search, Digital libraries, Memory aids, Neurodivergent support, Visualization

ACM Reference Format:

Anita Seihoon, Lynne Beveridge, and Orland Hoeber. 2026. SearchPath: A Visual Reminder of Past Search Activity to Support Students with Dyslexia when Undertaking Exploratory Searches. In *2026 ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR '26)*, March 22–26,

2026, Seattle, WA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3786304.3787926>

1 Introduction

Undertaking complex search tasks in academic environments is challenging [11]. The complexity of the task is a result of multiple factors, including being ambiguous, multi-faceted, or open-ended [25]. However, the types of search behaviours that people are accustomed to using for common look-up tasks are not well suited to undertaking complex search tasks. Instead, the complexity of the task calls for more complex search behaviour.

Exploratory search is a framework for searching that is well-suited to complex search tasks [10, 25]. The exploratory search process consists of two types of searching: an exploratory browsing style of searching where heuristic strategies that enable discovery, learning, and investigation are critical; and a focused style of searching, where analytical strategies consisting of query reformulation, result examination, and information extraction are undertaken. An underlying factor in this model is the concept of uncertainty. When a searcher begins their complex search task, their uncertainty will be high; by undertaking multiple rounds of exploratory browsing, knowledge is gained that reduces the uncertainty, which in turn enables the searcher to transition to focused searching on specific facets of the search task. Uncertainty can rebound as a result of discovering new aspects of the task, leading to a return to exploratory browsing and further searching to reduce the uncertainty [10, 13]. A common outcome of undertaking an exploratory search is that many queries will be issued and many search results will be saved along the way to a resolution of the search task.

Considering the context of academic search, without explicit support built into the search interface, university students find it challenging to match their search strategy to the complexity of the search task [11]. To address this problem, a variety of approaches have been proposed in the recent years to help searchers undertake and manage their exploratory search processes when faced with complex search tasks [8]. These generally fall into two categories: those that help searchers to make sense of the search results provided on the search engine results page (SERP) [6, 10, 14, 16, 23]; and those that use workspaces to help searchers manage their exploratory search activities [3, 6, 9, 10, 19].

The design and study of such interfaces within an academic search context has generally focused on students, with an underlying assumption that they will not have difficulties reading and understanding textual descriptions of search results, have the working memory capacity for monitoring progress towards a search goal, and are able to maintain focus on the search task. These assumptions can marginalize searchers with cognitive differences



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ACM ISBN 979-8-4007-2414-5/26/03

<https://doi.org/10.1145/3786304.3787926>

such as dyslexia [15], who may face challenges in processing dense academic texts, tracking search progress, and maintaining attention [20]. Few have considered their proposed approaches for supporting exploratory search in the context of accessibility.

In order to advance the field of interactive information retrieval, it is important that consideration be given to how the approaches developed are accessible when searchers are neurodiverse with respect to cognitive functions [15]. In this research, our focus is on supporting university students with dyslexia as they undertake exploratory search activities. Following a design-oriented research approach [5], we have drawn inspiration from recent advancements in exploratory search interface design including a set of design principles for exploratory search interfaces [8], leading to the development of SearchPath.

The fundamental goals of SearchPath are (1) to provide a multi-workspace environment that allows searchers to manage multiple concurrent search interests; (2) when a searcher saves a search result, provide a lightweight mechanism for adding information on why it is being saved; (3) to integrate an interactive memory aid into the SERP to help searchers to maintain orientation as they issue multiple queries and assess many search results throughout an exploratory search process; and (4) to provide a filtering mechanism within the saved search results list that reduces the cognitive load associated with analyzing what has been saved.

2 Literature Review

Dyslexia is a multi-dimensional type of neurodiversity without a single universally accepted definition. It commonly manifests as challenges with processing textual information, limitations in working memory, attention, and concentration, and low self-efficacy [20, 21]. The impacts of these challenges are particularly acute when conducting information search tasks [1, 2, 4], particularly in tasks involving reading, memory, and terminology [7].

Academic search tools often assume high reading fluency and verbal working memory, which can marginalize users with dyslexia [15]. Studies show that searchers with dyslexia struggle with spelling as they craft queries, skim dense result lists rather than reviewing them carefully and methodically, and have difficulties recalling which documents they previously accessed [17]. As searching within digital libraries is a fundamental academic activity [11] that has been shown to be an important factor in student success [22], it is vital to support students' neurodivergent cognitive functions such as dyslexia, in undertaking academic search tasks [18].

Research on the of pursuit complex and open-ended search tasks among undergraduate students with dyslexia highlights the challenges they face in keeping track of queries they have issued and which results they have examined [4]. Follow-up research emphasized their struggle to interpret predominantly textual search results, resulting in low self-efficacy and feelings of uncertainty during the search process [2]. Beyond the scope of searching, students with dyslexia experience more cognitive lapses, which manifest as being disorganized and prone to distractions [12].

The design of search interfaces to support exploratory search has become a rather active research topic in recent years [8]. While such exploratory search interfaces have been designed to support

a variety of search contexts, we focus our attention on those developed to enable searching among academic resources (academic digital libraries or academic repositories). We group this body of research into two general categories: approaches that enhance the SERP and approaches for monitoring or managing what was found.

Enhancements to the SERP in support of exploratory search typically employ either mechanisms for enabling query refinement or approaches for streamlining the evaluation of search results. Query refinement is supported via query suggestions; moving beyond simple query-completion approaches, some have studied how providing visual indications of the value of the query suggestion can benefit the searcher as they choose among the proposed queries [16]. Mechanisms for supporting the evaluation of the search results are more varied, including keyword highlighting [6, 10], using histograms and colour encoding [14], and generating concept maps [23]. Generally, such approaches add visual and interactive complexity to the search interface, under the premise that the cognitive complexity in using the interface is balanced with the ability for the searcher to produce better search outcomes for their complex search tasks.

Research on how to support searchers in managing what has been found generally focus on the development of workspaces. These can be single system-level workspaces for storing saved search results, providing interactive mechanisms for revealing relationships among saved search results [10], filtering features to reduce the complexity of the information space [19], or interactive controls for dynamically organizing saved search results [3]. Multi-workspace environments have also been proposed to help searchers keep track of their various academic search interests [6]. In such environments, the searcher enters the workspace before they start searching, resulting in the workspace encapsulating both the SERP and the collection of saved search results.

While most approaches to saving search results use a simple "save" button, some have studied approaches for tagging search results as they are saved [19]. Doing so serves as a mechanism for enabling the searcher to add brief explanations for why search results were found relevant or important. Beyond workspace-level support, some have studied the use of timelines added to the SERP to enable the monitoring of search activity and progress [9, 27].

A gap exists in the literature with respect to the design of exploratory search interfaces specifically for those with neurodiverse cognitive functions such as visual processing of text, working memory, and attention. As studies have found that many people with dyslexia prefer visual information while searching [1, 4], we hypothesize that academic digital library search interfaces that present visual and interactive features designed to support exploratory search will better support the search needs of people with dyslexia than traditional library search interfaces. This paper presents a first step towards testing this hypothesis.

3 SearchPath

SearchPath is a multi-workspace academic search interface, assisting students in keeping the searches for their various different academic interests separate from one another [6]. Before a search can be started or resumed, a workspace must be created or selected.

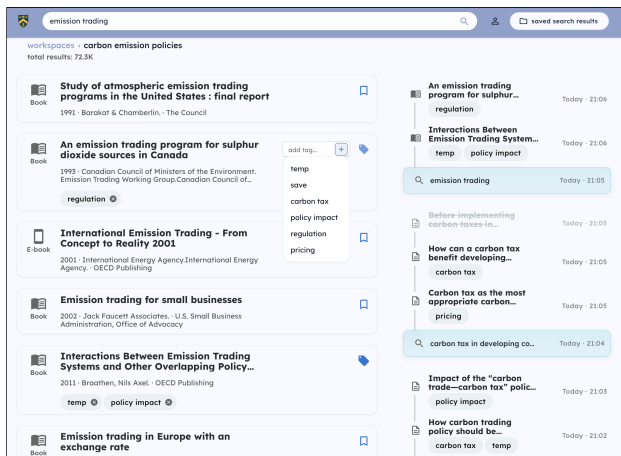


Figure 1: The SERP provides a timeline of the search activity, and an ability to tag search results as they are saved.

To assist searchers who require additional working memory support (such as those with dyslexia), workspaces can be personalized with the use of colour, serving as mechanism for maintaining awareness of which one was selected. This colour is used when viewing the list of all workspaces, and the ambient colour of the search interface when the workspace is selected. Due to limits of colour perception [24], this option is limited to a set of eight low-saturation colours that are perceptually distinct; searchers have the option to assign the same colour to related workspaces (e.g., those associated with the same course they are taking). Since the searchers are the ones who choose among these colours when personalizing their workspaces, those with deficiencies in colour perception will be able to choose colours that they personally perceive as distinct.

When a workspace is selected, all searching from that point forward is explicitly associated with that workspace. In addition to the ambient colour of the interface changing based on which workspace was selected, the title of the workspace is provided under the query box to remind the searcher of what they were doing if they happen to become distracted.

After issuing a query, searchers are able to save relevant search results to the “Saved Search Results” section of the current workspace. A typical approach to doing so is to provide a simple save button or icon beside each search result. However, there is a risk that this can result in searchers losing track of why particular search results were saved, especially when pursuing complex search tasks that are multi-faceted. Drawing inspiration from prior work on tagging search results [19], SearchPath allows searchers to add short text-based tags as part of the save process. Studies of this approach have shown that this approach fosters user engagement while undertaking complex search tasks, as well as increases searchers’ perceived motivation and ability during the search process [19]; we hypothesize that the agency in choosing tags to explain why search results were saved may assist students with dyslexia in mitigating their working memory neurodiversity.

When searchers select the save icon for a given search result, they are provided with a textbox in which they can add one or more tags, creating a flexible environment for documenting why a particular

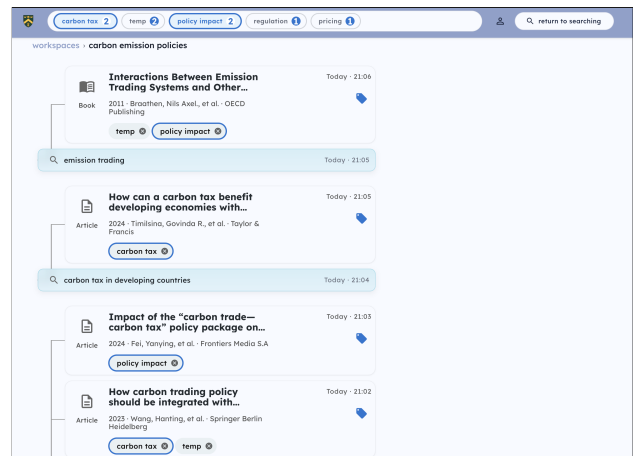


Figure 2: The saved search results provides a mechanism for filtering the search results based on the searcher’s tags.

search result is relevant or interesting. To reduce the cognitive load of having to remember or correctly type previously used tags, an auto-completion mechanism is provided; as the searcher types, tags are suggested that were previously used within the current workspace. Doing so reduces the need for the searcher to have to recall and re-type tags they previously created.

While this tagging approach adds cognitive effort when saving a search result, it allows the searcher to be thoughtful about why they are saving a given search result. As searchers with dyslexia and others with neurodiverse language function may have difficulty choosing terms to use, SearchPath provides two default tags that can easily be selected from the auto-completion menu: “temp” and “save”. These tags provide searchers with a starting point for tagging saved search results (temporarily saved and saved without any extra explanation), reducing the decision-making needed to save a search result early in the search process when the searcher may not yet know how to effectively use the tags. They can return to the saved search result later to provide more descriptive tags as their uncertainty in the exploratory search activity is reduced.

As queries are issued and search results are saved, these activities are presented within the SERP as a timeline visualization of the search history. This is inspired by previous work, where providing an overview timeline on the SERP and a detailed timeline of saved search results in the workspace resulted in clear usability, user engagement, and perceived knowledge gain benefits in the context of cross-session exploratory search [9]. We hypothesize that the value provided for cross-session task reacquaintance translates to value for searchers with neurodivergent working memory and attention functions.

This timeline serves as an external artifact that documents the searcher’s activities within the given workspace over all search sessions, including queries issued, search results saved, and the tags added to those search results. Such externalizations serve as memory aids, providing a visual representation of their search path. By providing this timeline within the SERP, searchers can reference it at any time during their search activity, helping them to stay focused while offloading demands on their working memory.

Further, it provides a mechanism that can allow the searcher to connect their current search activity to what was done previously, avoiding repetition of past search activities. Figure 1 shows this timeline in the midst of an exploratory search, as well as the tagging mechanism when a search result is selected for saving. Here, the ambient colour has been personalized by the searcher to be blue; the name of the selected workspace is also provided to remind the searcher of their task.

At any point in time, the searcher can switch from searching to viewing the search results that have been saved in the workspace (see Figure 2). These saved search results are presented in the same format as used in the SERP, including how the searcher-supplied tags are represented. At the top of the saved search results page, each unique tag is displayed along with a count of how many times it was used in the workspace. These tags are interactive; by selecting them the search results are filtered to only include those that use these tags. Selecting multiple tags in this manner is treated as a logical OR, including search results that use any of the selected tags. This filtering serves as a mechanism for helping to reduce the information overload when reviewing the saved search results. Further, as the tags may be used to represent different aspects of the task, it assists searchers in managing the multi-faceted aspect of complex search tasks.

At any time, the searcher can select a saved search result and make changes to the tags assigned to it, including deleting tags and adding new ones. This works the same in the list of saved search results as it does in the SERP. If all the tags for a saved search result are deleted, the search result is removed from the set of saved search results. However, it is preserved in the timeline but dimmed and with a strike-out font styling, serving as a history of what was considered relevant but later discarded.

Much of the work on accessible design for users with dyslexia focuses on the readability of text [28]. Consistent with general guides for web usability [26], careful consideration should be given to the font choice. Although a general rule-of-thumb is that a sans-serif font is more readable in electronic format than a serif font, this is not universally true. As some sans-serif fonts provide only small visual differences between certain characters, a more thoughtful approach to choosing the fonts in an interface for users with dyslexia is needed. We chose Lexend Deca, which has been designed to improve readability by increasing spacing between characters.

Careful consideration was given to how guidelines written in the context of the readability of blocks of text can be applied to the textual elements in a search interface. Consistent with these guidelines, the titles of the search results are 20% larger than the normal text, with a high contrast between foreground and background (while avoiding black and white), and with less than 60 characters per line. However, the metadata for the search results (document type, year, author, etc.) are represented in a lighter colour at the normal font size so as to appear visually different than the title and not signal to the searcher that it requires the same level of scrutiny when considering relevance. While the guidelines suggest line spacing to be 150% of the font height, we instead chose to keep it at the default 100% in order to keep the search result cards and timeline more compact. However, to further reinforce the difference between the title of a search result, the metadata about it, and the user-supplied

tags, the line spacing between these elements is 120% of the title font height, creating a noticeable gap between each element.

Similarly, the line spacing in the tag auto-completion list is smaller than the guideline as any multi-line tags are ones that the searcher typed; the line spacing between elements is set at the recommended 200%. The icons representing the different search result document types were chosen to appear visually unique; their meaning is reinforced with labeling in the search results list.

SearchPath's features were designed in consideration of recently published design principles for exploratory search [8]. In particular, all the features have been designed to be lightweight additions to existing search interface design patterns. Wherever possible, visual representations are provided either in place of textual representations or to supplement the text. Such visualizations have been designed to be scrutable, enabling the searcher to make sense of what is being shown in relation to their search activity. There is a high degree of interactivity provided with these features, which is persistent within and between search sessions.

4 Implementation Details and Demo Video

SearchPath has been implemented as a web-based application using the MERN stack (MongoDB, Express.js, React, Node.js) and using the Material Design 3 UI framework. The interface integrates with our university's Ex Libris Primo API, providing near real-time search results that are identical to those provided to searchers of our library catalog. All search activities are saved in a no-SQL database, providing the data source for the generation of the timeline and saved search results list. A demo video is available on the ACM Digital Library record for this paper.

5 Discussion

The key benefit of SearchPath is that it provides scaffolding for offloading working memory during an exploratory search process. This is done automatically as searchers issue queries and save search results, as well as interactively as saved search results are tagged. There is no need for the searcher to remember what they have done, nor why they have chosen to save specific search results, as this information can easily be viewed in the timeline visualization of the search activity. By offloading working memory, this limited resource is freed up for other important work, such as reading and considering the relevance of individual search results. For students with dyslexia, whose working memory is consumed by reading [20], any approach that reduces other working memory demands is expected to be beneficial.

The secondary benefit of SearchPath is the support provided for those with lapses in attention. The timeline added to the SERP is expected to reduce the cognitive load of resuming the search when the lapse is short. If the lapse is longer, the support for task resumption is provided through the interactive features in the saved search results (e.g., viewing queries issued and what was saved, reconsidering the tags used, filtering the saved search results to assess subsets of the saved results).

Search Path is presented here as a demonstration of a set of approaches that have been designed to support neurodiverse student searchers with respect to reading, working memory, and attention. Our next step is to design and conduct a user study with a specific

population of neurodiverse students: those with dyslexia. Such a study will not only consider the typical search interface metrics, but will also assess aspects of uncertainty and self-efficacy, which have been shown to be important factors for students with dyslexia [2].

Acknowledgments

The authors acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC), through the Alliance Grant held by the last author (ALLRP 558319 - 20).

The authors would like to thank Stephann Makri and Andrew MacFarlane for their contributions to this research. We appreciate the time they devoted to discussing early conceptualizations of this research, and to review draft writing.

This research was conducted at the University of Regina, which is situated on Treaty 4 and Treaty 6 territories - the homelands of the nêhiyawak, Anihšināpēk, Dakota, Lakota, Nakoda peoples, and Michif/Métis nation.

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